PP-S-31F
May 26, 1976
SUPERSEDING
Fed. Spec. PP-S-31E
March 27, 1967

FEDERAL SPECIFICATION

SALMON, CANNED

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers the requirements for the method of preparation, packaging and packing of canned salmon for use by agencies of the Federal Government.

1.2 Classification.

1.2.1 Styles, forms, and species. Canned salmon shall be of the following styles, forms, and species, as specified (see 6.1).

Style 1 - Skin and bones included
Style 2 - Skinned and backbone removed
Form a - Regular
Form b - No salt added. The sodium content shall be not more than 60 mg. per 100 grams (g.) of drained flesh.

Species:

(a) Chinook, king or spring (Oncorhynchus tshawytscha)
(b) Red, sockeye or blue back (Oncorhynchus nerka)
(c) Silver, coho, cohoe, medium or red (Oncorhynchus kisutch)
(d) Pink (Oncorhynchus gorbuscha)
(e) Chum or keta (Oncorhynchus keta)

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

PPP-C-29 - Canned Subsistence Items, Packaging and Packing Of

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(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

(Single copies of this specification and other Federal specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.)

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-1483 - Sanitary Standards for Fish Plants

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity of as directed by the contracting officer.)

LAWS AND REGULATIONS

U. S. Department of Health, Education and Welfare

Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

U. S. Department of Commerce

Regulations Governing Processed Fishery Products

(Application for copies should be addressed to the U. S. Department of Commerce, National Oceanic and Atmospheric Administration, National Marine Fisheries Service, Office of Inspection, Washington, DC 20240.)

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U. S. Department of Agriculture

U. S. Standards for Condition of Food Containers

(Application for copies should be addressed to the Information Division, Agricultural Marketing Service, U. S. Department of Agriculture, Washington, DC 20250.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply:

Association of Official Analytical Chemists

Official Methods of Analysis of the Association of Official Analytical Chemists

(Application for copies should be addressed to the Association of Official Analytical Chemists, P. O. Box 540, Benjamin Franklin Station, Washington, DC 20044.)

3. REQUIREMENTS

3.1 Materials.

3.1.1 At the time of processing, the salmon shall be fresh, of good quality, good odor, free of foreign material, heads, viscera, fins, blood clots, and shall have been washed. Hatchery salmon (salmon after spawning) shall be removed and excluded from delivery. The canned salmon shall be prepared from species listed in 1.2 to produce a finished product meeting requirements of 3.4.

3.1.2 Salt (sodium-chloride). Form "a" only, shall be of edible grade with or without anticaking agents. Iodized salt shall not be used.

3.1.3 Oil. No oil other than natural salmon oil may be added. If natural oil is added, it shall be comparable in color, viscosity, and all other chemical requirements to that oil which would occur in the product when processed without added natural oil.

3.2 The salmon and oil shall conform to the following characteristics:

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TABLE I. Product characteristics

Species	Color of oil	Color of flesh	Texture of fish
Chinook or king (spring)	Light red through orange to almost white	Red to orange red to light pink	Moderately firm to soft
Red or sockeye (blueback)	Deep red to deep orange	Dark red to orange red	Very firm to soft
Silver, coho, cohoe (medium red)	Light red to yellowish pink	Red to orange red to light pink	Moderately firm to soft
Pink	Pink to light yellow	Pink to buff	Moderately firm to soft
Chum or keta	Light pink with orange shade to yellow	Light pink to grayish white	Moderately firm to firm

3.3 Processing. The product shall be processed from one species and other applicable component materials specified in 3.1 to produce a product complying with 3.4. In addition, the container shall be filled, vacummized, sealed, and heat-treated in such a manner as will stabilize the product for safe use by Government agencies where refrigerated supply conditions are not available. Style 1 shall be prepared with skin and bones included. Style 2 shall be skinned and backbone removed. Form a, shall be regular canned salmon with salt added. Form b, shall be canned salmon with no salt added; the sodium content shall be limited to a maximum of 60 mg. per 100 g. of drained weight of flesh.

3.4 Finished product. The finished product shall comply with the requirements of table II.

TABLE II. Physical requirements

Item No.	Requirement
1	The product shall be free from fins, gills or viscera and shall contain only one species of salmon. Presence of skin or backbone will automatically classify the product as style 1.
2	The product shall be free from mutilated areas, blood clots or bruises, and hard bones (see 6.7.1, 6.7.2 and 6.7.3).
3.	The product shall be free of watermarking (see 6.7.4).

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TABLE II. Physical requirements

Item No.	Requirement
4	There shall be no more than three pieces and one patch portion (see 6.7.5) in cans of one pound (454 g) or smaller, nor more than 12 pieces and one patch portion in a 4-pound (1.81 kg) can (style 1 only).
5	The product shall be not more than 5 percent under the specified net weight (see 5.1.1).
6	The product shall be free of foreign material (see 6.7.6).
7	The product shall be free from objectionable flavor and odor. The product shall agree with the color and texture of flesh and color of oil listed in 3.2 for the species specified (see 6.1 and 6.7.7).
8	The product shall be free from honeycombed flesh (see 6.7.8).
9	A can shall have a minimum of 5 inches of vacuum except that the 3-3/4 ounce (106 g) cans shall have a minimum of 2 inches (94.2 kPa) of vacuum and 7-3/4 ounce (220 g) can shall have a minimum of 3 inches (90.8 kPa) of vacuum.
10	The product shall be packaged so that the cut surfaces are approximately parallel with the ends of the container.

3.5 Federal Food, Drug, and Cosmetic Act. All deliveries shall conform in every respect to the provisions of the Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder.

3.6 Plant qualification. The product shall be prepared, handled, and delivered under the sanitary conditions set forth in applicable paragraphs contained in Part 260 of the National Oceanic and Atmospheric Administration Regulations Governing Processed Fishery Products, Code Title 50, and shall be listed in the latest issue of the Approved List of Sanitarily Inspected Fish Establishments. For Military procurement, when the above regulations are not enforced, the product shall be processed in establishments meeting the requirements of MIL-STD-1483.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Preacceptance inspection (plant sanitation inspection).

4.2.1 For civil agencies. When specified (see 6.1), inspection of plants producing the product will be conducted upon request of the contracting officer to the appropriate Government inspection agency. Award of contracts will be limited to establishments known to maintain proper sanitary conditions or which have received prior sanitary approval recommended by the inspection agency.

4.2.2 For military agencies. Product shall be rejected if produced in plants not meeting the requirements of 3.6.

4.3 Acceptance inspection.

4.3.1 Condition inspection of containers. When inspection of product for quality and condition is performed by the U.S. Department of Commerce, the United States Standards for Condition of Food Containers shall apply. (For applicable criteria governing inspection of containers for the military, see 4.3.4.1).

4.3.2 Sampling for inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.3.3 Component and material inspection. In accordance with 4.1 above, components and materials shall be inspected in accordance with all the requirements of referenced specifications, drawings and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase documents.

4.3.3.1 Process examination. Examination shall be made during processing to determine the requirements of 3.3. Records of time and temperature shall be maintained. Nonconformance to one or more of the above referenced requirements reflected by actual examination or by records shall be cause for rejection of the lot or finished product made therefrom, as applicable.

4.3.4 Examination of finished product. Classification of defects found during examination shall be in accordance with 4.3.4.1 and tables IV through VI. The inspection levels, sample units, expression of lot sizes, and the acceptable quality levels (AQL's) for tables IV through VI shall be as shown below. AQL's shall be expressed in defects per hundred units.

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TABLE III. End item examination criteria

Table 1/	Sample unit	Lot size expressed in	Inspection level 2/	AQL's Major B	AQL's Minor
IV	Contents of one primary container	Shipping container	I	---	6.5
V	Contents of one primary container	Shipping container	I	---	6.5
VI	Contents of one primary container	Shipping container	I	1.5	6.5

1/ Applicable to 3-3/4 ounce (106 g), 7-3/4 ounce (220 g), 15-1/2 ounce (439 g) or 4-pound (1.81 kg) cans.

2/ For military procurement, sample shipping containers shall be selected randomly on a proportionate basis from not less than 90 percent of the code listed on the warehouse receipt. In the event that the referenced levels result in a sample size that is too small to provide for sampling 90 percent of the codes on the warehouse receipt, a level sufficiently large to accommodate this requirement shall be selected. Not more than one primary container per shipping container shall be examined for the defect listed in table IV.

TABLE IV. Examination for net weight 1/ 2/

Category	Defect
Minor	
201	A single primary container with more than 5 percent under specified weight (see 5.1.1).

1/ Lot shall be rejected if sample data indicates that lot average net weight is less than specified net weight.

2/ Report results to the nearest 1/4 ounce (g).

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TABLE V. Examination of packaging properties

Categories	Defect
Minor	
201	Cross fill. 1/
202	More than three pieces and one patch portion in a 7-3/4 ounce (220 g) or 15-1/2 ounce (439 g) can only. More than 12 pieces and one patch portion in a 4-pound (1.81 kg) can (see 6.7.5).

1/ The vertebral column not generally parallel to the vertical axis of the can.

2/ Not applicable to style 2 salmon.

TABLE VI. Examination of product characteristics 1/

Categories			Defect
Major A	Major B	Minor	
101			Presence of objectionable flavor or odor (see 6.7.7).
102			Presence of foreign material (e.g., hair, dirt, insect, insect parts, paper, wood, glass or metal particle or any other extraneous material).
103			Presence of honey combing (see 6.7.8).
	151		Bones not soft and friable (see 6.7.3).
	152		Species of salmon not that specified.
	153		Presence of backbone (applicable to style 2).
		201	Presence of watermarking (see 6.7.4).
		202	Presence of gills, fins or viscera.
		203	Presence of skin (applicable to style 2).
		204	Color of flesh not characteristic of species. 2/

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TABLE VI. Examination of product characteristics 1/

Categories			Defect
Major A	Major B	Minor	
		205	Color of oil not characteristic of species. 2/
		206	Texture not characteristic of species. 2/
		207	Presence of blood clots, mutilated areas or bruises (see 6.7.1 or 6.7.2, as applicable).

1/ The finding of any Major A defect shall be basis for rejection of the lot.

2/ See 3.2 for characteristic texture and color of oil for applicable species.

4.3.4.1 Examination of packaging, packing, labeling, marking and can vacuum. Examination of packaging, packing, labeling, marking and can vacuum shall be in accordance with PPP-C-29 and subsidiary specification where therein directed.

4.3.5 Sampling procedures and acceptance criteria for testing of finished product. The finished product, form b, shall be tested for sodium content as specified in 3.3. Procedures for testing shall be in accordance with 4.3.6.1. Lot size shall be expressed in primary containers. The sample for testing shall be an 8 ounce (227 g) composite derived from the number of primary containers indicated by inspection level S-2. Results shall be reported in accordance with the applicable reference(s) in 4.3.6.1. The lot shall be rejected if test results indicate non-conformance with requirements.

4.3.6 Tests. Test procedures and controls which differ from those specified herein may be used by the supplier if they provide a quality assurance equivalent to that specified. If the Government contracting officer determines that such procedures and controls do not provide, as a minimum, such quality assurance, the contractor will use the test procedures set forth herein. In case of dispute as to the test results, the test methods specified herein will govern.

4.3.6.1 Chemical analyses. Unless otherwise specified (see 6.1), chemical analyses, if required by the purchasers, shall be in accordance with the Official Methods of Analysis of the Association of Official Analytical Chemists in effect on date of invitation to bid.

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5. PREPARATION FOR DELIVERY

5.1 Packaging and packing. Packaging shall be level A or C, packing shall be level A, B, or C, as specified (see 6.1).

5.1.1 Packaging, packing, labeling, and marking. The product shall be packaged, packed, labeled, and marked in accordance with PPP-C-29. The can size, net contents, and vacuum requirements shall be as specified in table VII.

TABLE VII. Net contents and vacuum requirements

Can size	Net contents Ounces	Vacuum requirements Inches
301 x 106	3-3/4 (106 g)	2 (94.2 kPa)
307 x 200.25	7-3/4 (220 g)	3 (90.8 kPa)
301 x 408	15-1/2 (439 g)	5 (84.08 kPa)
301 x 411	16 (454 g)	5 (84.08 kPa)
603 x 405	64 (1.8 kg)	5 (84.08 kPa)

6. NOTES

6.1 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Style, form, and species required (see 1.2).
(c) When inspection of plants is required for civil agencies (see 4.2.1).
(d) When chemical analyses are required (see 4.3.6.1).
(e) Selection of applicable levels of packaging and packing required (see 5.1).

6.2 Bid samples. If for any particular purpose samples with bids are necessary, they should be specifically asked for in the invitation for bids, and the particular purpose to be served by the bid sample should be definitely stated.

6.3 This specification does not include all styles or forms of canned salmon that are commercially available but is intended to cover the types, etc., which are suitable for Federal government requirements (see 1.2).

6.4 Supplier's inspection. The supplier shall be required to assure that the product conforms with the specification prior to submission to the Government for final inspection.

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6.5 Destination inspection (for civil agencies). When the finished product has been inspected and passed at point other than destination, the contract should require that the product be inspected at destination for identity, condition, and quantity only.

6.6 Award of contract for the product specified in this document will be limited to plants known to maintain the required sanitation conditions of 3.6 (also see 4.2).

6.7 Definition of terms.

6.7.1 Mutilated areas. Mutilated areas are areas of flesh which are not physically intact due to mechanical or predatory damage.

6.7.2 Blood clots or bruises. Blood clots or bruises are those which detract from the appearance of the product. Minute blood streaks or blood spots which do not materially affect the appearance are not considered a defect.

6.7.3 Hard bones. Hard bones are those which are not soft or friable. Bones are soft and friable if the vertebrae are easily crushed when pressed between thumb and fingers. The rib bones should be brittle and breakable with slight distorting finger pressure.

6.7.4 Watermarking. Watermarking is one or more of the following conditions: (1) the skin shown shades of red or yellow; (2) the normally white areas of the skin show mottled gray or black; (3) The skin shows vertical black stripes or irregular color patterns.

6.7.5 Patch portion. A patch portion is a small portion of salmon which is added, if necessary, to complete the fill of the container.

6.7.6 Foreign material. Foreign material is any material other than skin, bones, flesh, and natural fluids of properly eviscerated, cleaned and processed fish, and additional oil and salt as provided for in 3.1.2 and 3.1.3.

6.7.7 Objectionable flavor and odor. Objectionable flavor and odor includes deteriorative flavor and odor which occur in improperly captured or improperly preserved fish, abnormal odors, and flavors due to unwholesome conditions, or environmental odors and flavors resulting from contact with contaminating surfaces or substances. These shall include, but not be limited to, putrid, ammoniacal, rancid, tainted, sour, stale, scorched, excessive salt flavor, turpentine, kerosene, or fuel oil.

6.7.8 Honeycombed. Honeycombed is the visable appearance of discrete holes or openings of varying size throughout the salmon flesh.

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MILITARY CUSTODIANS:

Army - GL
Navy - SA
Air Force - 45

Review activities:

Army - MD
Navy - MC, MS
DP - SS

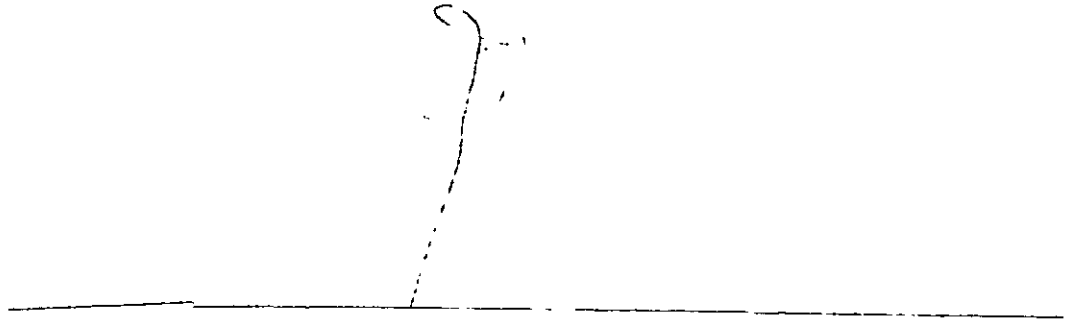
Civil Agency Coordinating Activities:

GSA - FSS

Preparing activity:

Army - GL

Project No. 8905-0793



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